

Video Game Film Adaptations — Factors in Box Office and Critical Reception

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Introduction & Motivation

Video game film adaptations occupy a unique and often-maligned corner of the entertainment industry. For decades, translating interactive gaming experiences to the passive medium of cinema has proven notoriously difficult, producing a catalogue of films that range from critically panned box-office bombs to billion-dollar cultural phenomena. The 2023 release of *The Super Mario Bros. Movie*, which grossed over \$1.3 billion worldwide, demonstrated that video game adaptations can achieve extraordinary commercial success when the right combination of intellectual property, creative vision, and audience goodwill aligns.

Yet the genre’s history is littered with failures: early attempts like *Super Mario Bros.* (1993) and *Street Fighter* (1994) became cautionary tales, while more recent entries like *Monster Hunter* (2020) and the *Resident Evil* reboot (2021) continue to underwhelm. This uneven track record raises important questions about what drives success in video game film adaptations. Do critics and audiences see eye to eye on these films? Does the financial muscle of the game publisher behind the IP translate to bigger box office returns? And do film franchises based on video games suffer from the same diminishing returns — “franchise fatigue” — that plagues many traditional Hollywood series?

In this project, we analyze a comprehensive dataset of video game film adaptations spanning nearly four decades to explore three core questions about the intersection of critical reception, publisher economics, and franchise longevity. Understanding these relationships may offer insight not only for film studios and game publishers seeking to develop successful adaptations, but also for audiences seeking to understand why some video game movies thrive while others flounder.

Research Questions

Our investigation is organized around three primary research questions:

1. **Critic–Audience Agreement Over Time:** Do critics and audiences agree on the quality of video game films, and has the gap between their assessments changed over time?
2. **Publisher Size and Box Office Performance:** Does the size of a game’s publishing company predict its film adaptation’s box office success?
3. **Franchise Fatigue:** Do video game film franchises exhibit “franchise fatigue” — declining box office returns and critical reception with successive installments?

Data Provenance

Primary Dataset

Our primary data source is the TidyTuesday *Game Films* dataset (June 9, 2026), hosted on the `rfordatascience/tidytuesday` GitHub repository. This dataset was originally scraped from Wikipedia’s “List of films based on video games” and compiled for the TidyTuesday weekly data exploration project. It contains observations on film adaptations of video games, with each row representing one film adaptation.

The raw dataset comprises 439 rows and 19 variables including: title, release date, category (theatrical, direct-to-video, television, etc.), original game publisher, worldwide box office with currency designation, budget bounds, Rotten Tomatoes critic score, Metacritic critic score, CinemaScore audience grade, and distributor information.

Supplementary Data

We supplemented the primary dataset with two external data sources:

- **CPI-U Inflation Data:** U.S. Bureau of Labor Statistics Consumer Price Index for All Urban Consumers (series CUUR0000SA0), annual averages from 1986 through 2024. These values were used to adjust all monetary figures to constant 2024 dollars, enabling fair comparison of box office performance across decades.
- **Publisher Revenue Data:** Wikipedia’s “List of largest video game companies by revenue,” scraped via the `rvest` package. This table provides estimated annual revenue for major video game publishers, which we used as a proxy for publisher size.

Data Wrangling & Cleaning

To prepare the dataset for analysis, we performed several cleaning and transformation steps. We describe each step narratively below; the corresponding code is available in the Code Appendix.

Variable Selection

We retained all original variables, as each played a role in at least one of our research questions. Key variables include `title` for franchise detection, `release_date` for temporal analysis, `original_game_publisher` for the publisher-size question, `worldwide_box_office` and `budget_low/budget_high` for financial analysis, and `rotten_tomatoes`, `metacritic`, and `cinema_score` for the critic-audience comparison.

Currency Standardization

The raw dataset contains monetary values in their original currency, indicated by a companion currency column. We standardized all financial figures to USD using the approximate exchange rates shown below:

Table 1: Exchange rates used for currency standardization

Currency	USD per Unit
\$	1.0000
¥	0.0091
£	1.3000
HK\$	0.1280
NT\$	0.0330

We joined this conversion table to the main dataset on both `worldwide_box_office_currency` and `budget_currency`, creating new USD-denominated columns: `worldwide_box_office_usd`, `budget_low_usd`, and `budget_high_usd`.

Inflation Adjustment

Comparing box office figures across nearly four decades requires adjusting for inflation. We joined a CPI-U table to the dataset, matching on the year extracted from `release_date`, and multiplied all USD amounts by the factor $\text{CPI}_{2024} / \text{CPI}_{\text{year}}$. This converts all monetary figures to constant 2024 dollars, ensuring that a dollar earned in 1993 is weighted equivalently to a dollar earned in 2024.

Film Category Filtering

The dataset includes films across several distribution categories, including direct-to-video releases and television specials, which often lack comprehensive financial or review data. We

filtered to retain only theatrical releases, as these represent the primary market for video game film adaptations and have the most complete data.

CinemaScore Conversion

The `cinema_score` variable is stored as a character column containing letter grades (A+ through F) and "N/A" for films without CinemaScore data. For quantitative analysis, we converted these letter grades to a numeric 0–100 scale using the mapping shown below. Films with "N/A" CinemaScores were assigned NA and excluded from analyses requiring audience scores.

Final Dataset Overview

After all cleaning steps, the dataset contains only theatrical-release video game film adaptations with standardized USD financial data adjusted for inflation to 2024 dollars, and a numeric cinema score. The summary below characterizes the final working dataset:

Table 2: Overview of the cleaned dataset after all preparation steps

Total Films	Films with Box Office Data	Films with RT Score	Films with Metacritic	Films with CinemaScore
246	89	73	63	51

Exploratory Data Analysis

To derive meaningful insights and address our three research questions, we conducted extensive exploratory data analysis. This section presents our key tables and figures, organized by research question. Each figure is numbered, captioned, and discussed in the narrative that follows.

Descriptive Statistics

Table 3 provides an overview of the key numeric variables in our dataset. The summary statistics (mean, median, minimum, maximum) offer context about the range and distribution of these variables before we examine their relationships.

Warning: Expected 2 pieces. Additional pieces discarded in 20 rows [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 21, 22, 23, 24, 25].

Warning: Values from `value` are not uniquely identified; output will contain list-cols.

* Use `values\_fn = list` to suppress this warning.

* Use `values\_fn = {summary\_fun}` to summarise duplicates.

* Use the following dplyr code to identify duplicates.

```
{data} |>
  dplyr::summarise(n = dplyr::n(), .by = c(Variable, Statistic)) |>
  dplyr::filter(n > 1L)
```

Variable	box
worldwide	1.525276e+08, 6.364336e+07, 2.085890e+08, 6.332787e+04, 1.400977e+09
budget	NULL 81431922, 70412468,
rotten	NULL
Metacritic	NULL
cinema	NULL

The distribution of box office values spans several orders of magnitude, from near-zero to over a billion dollars (inflation-adjusted), suggesting a heavily right-skewed distribution typical of entertainment industry financial data. Critic scores (Rotten Tomatoes and Metacritic) average in the 40–50 range, indicating that video game films tend to receive lukewarm-to-negative critical reception. CinemaScore audience grades average around 60 on our 0–100 scale — notably higher than critic scores, hinting at the critic–audience gap we investigate in Q1.

Q1: Critic–Audience Agreement Over Time

Our first research question asks whether critics and audiences agree on the quality of video game films and whether any gap between their assessments has changed over the nearly four-decade span of our dataset. To address this, we compared three score types: Rotten Tomatoes (critic aggregate), Metacritic (critic aggregate), and CinemaScore (audience survey, converted to numeric).

Figure 1 plots all three score types against release date, with loess-smoothed trend lines and 95% confidence bands. This allows us to assess both the overall trajectory of scores and the relative positioning of critic versus audience evaluations.

```
`geom_smooth()` using formula = 'y ~ x'
```

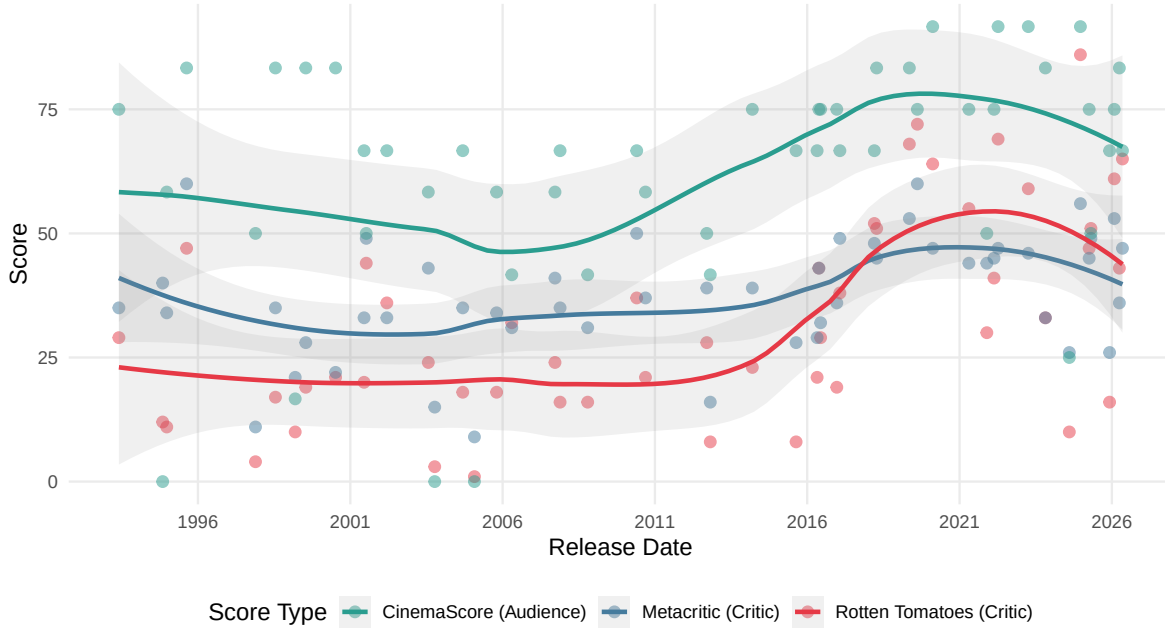


Figure 1: Critic and audience scores over time for video game film adaptations, 1986–2024. Loess-smoothed trends with 95% confidence bands.

Several patterns emerge from Figure 1. First, CinemaScore (audience) ratings consistently outpace both critic metrics across the entire timeline. The loess-smoothed CinemaScore trend hovers between 60 and 70 throughout, while Rotten Tomatoes and Metacritic trends cluster in the 40–55 range. This confirms a persistent critic–audience gap: audiences consistently rate video game films 10–20 points higher than critics do. Second, all three score types show a modest upward trend since approximately 2010, possibly reflecting improved production quality as studios have invested more seriously in video game adaptations following early failures.

To quantify the critic–audience gap directly, Figure 2 plots the difference between average critic scores (mean of Rotten Tomatoes and Metacritic) and CinemaScore audience ratings for each film. Points above the zero reference line indicate films where critics rated the film higher than audiences; points below indicate the opposite.

```
`geom_smooth()` using formula = 'y ~ x'
```

Figure 2 reveals that the overwhelming majority of video game films fall below the zero line — audiences rate these films higher than critics. The loess-smoothed trend suggests a slight narrowing of the gap over time, particularly after 2010, though the confidence band widens in recent years due to sparser data. The persistence of negative gaps across four decades

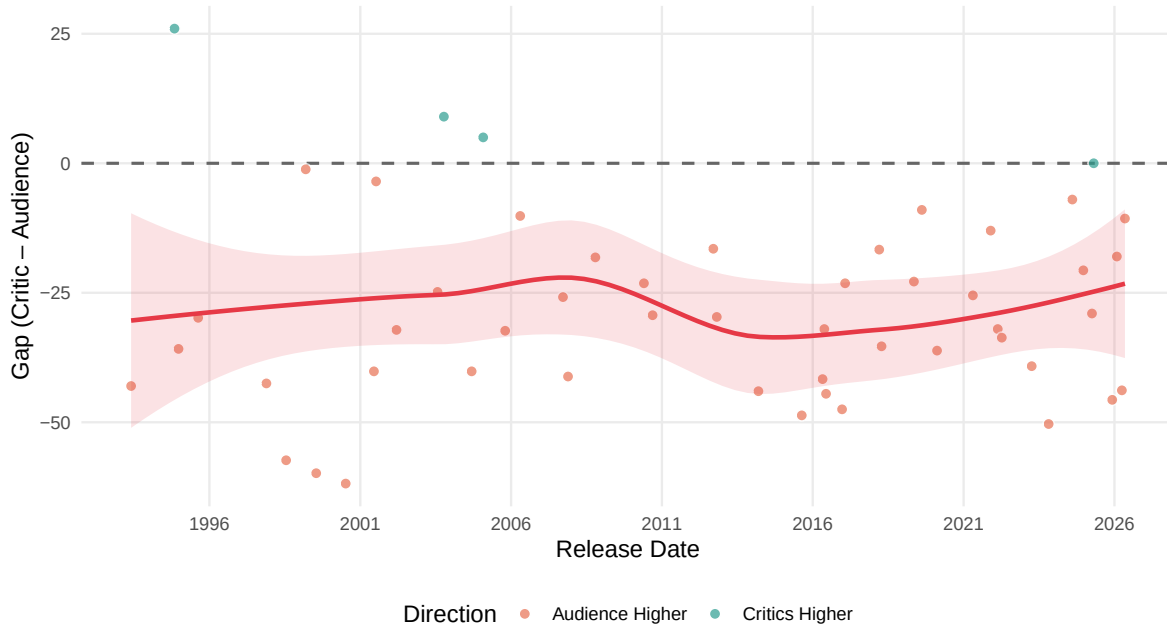


Figure 2: Critic–audience score gap over time. Positive values indicate critics rated films higher than audiences; negative values indicate the reverse.

underscores a fundamental divergence between how professional critics and paying audiences evaluate video game film adaptations.

Key findings for Q1: Audiences consistently rate video game films higher than critics, with a gap of roughly 10–20 points that has narrowed modestly since 2010. The overall upward trend in all score types since 2010 suggests improving quality in the genre.

Q2: Publisher Size and Box Office Performance

Our second research question investigates whether larger video game publishers — measured by annual corporate revenue — produce film adaptations that earn more at the box office. Intuitively, one might expect that companies with greater financial resources (like Nintendo or Sony) would invest more heavily in film adaptations of their intellectual property, leading to higher production values and stronger marketing campaigns that translate to greater box office returns.

To test this hypothesis, we scraped Wikipedia’s list of the largest video game companies by annual revenue and joined it to our film dataset by publisher name. After filtering to

matched entries with complete data, we retained 36 films spanning 8 distinct publishers for this analysis.

Table 4 presents aggregated performance metrics for publishers with at least three films in the matched dataset, ranked by mean inflation-adjusted box office.

Table 4: Publisher Performance — Aggregated Box Office and Critic Me
3+ matched films)

company	# Films	Median Box Office (M USD)	Mean Box Offi
Sega	4	411.0	
Eidos	3	266.8	
Ubisoft	4	157.8	
Capcom	15	51.1	
Square Enix	4	58.8	
Nintendo | The Pokémon Company	24	52.3	
Type-Moon	3	24.3	

Figure 3 is our primary multivariate visualization for this question, encoding four variables in a single scatterplot: publisher revenue (x-axis, log scale), inflation-adjusted box office (y-axis, log scale), publisher identity (color), and film budget (point size). This figure satisfies the requirement to display three or more variables in a single visualization.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

```
`geom_smooth()` using formula = 'y ~ x'
```

Warning: The following aesthetics were dropped during statistical transformation: size.
i This can happen when ggplot fails to infer the correct grouping structure in the data.
i Did you forget to specify a `group` aesthetic or to convert a numerical variable into a factor?

Warning: Removed 7 rows containing missing values or values outside the scale range (`geom\_point()`).

The dashed linear-model trend line in Figure 3 shows only a very weak positive association between publisher revenue and box office earnings. While the two highest-grossing films in our sample (both *Super Mario* adaptations from Nintendo) correspond to one of the largest

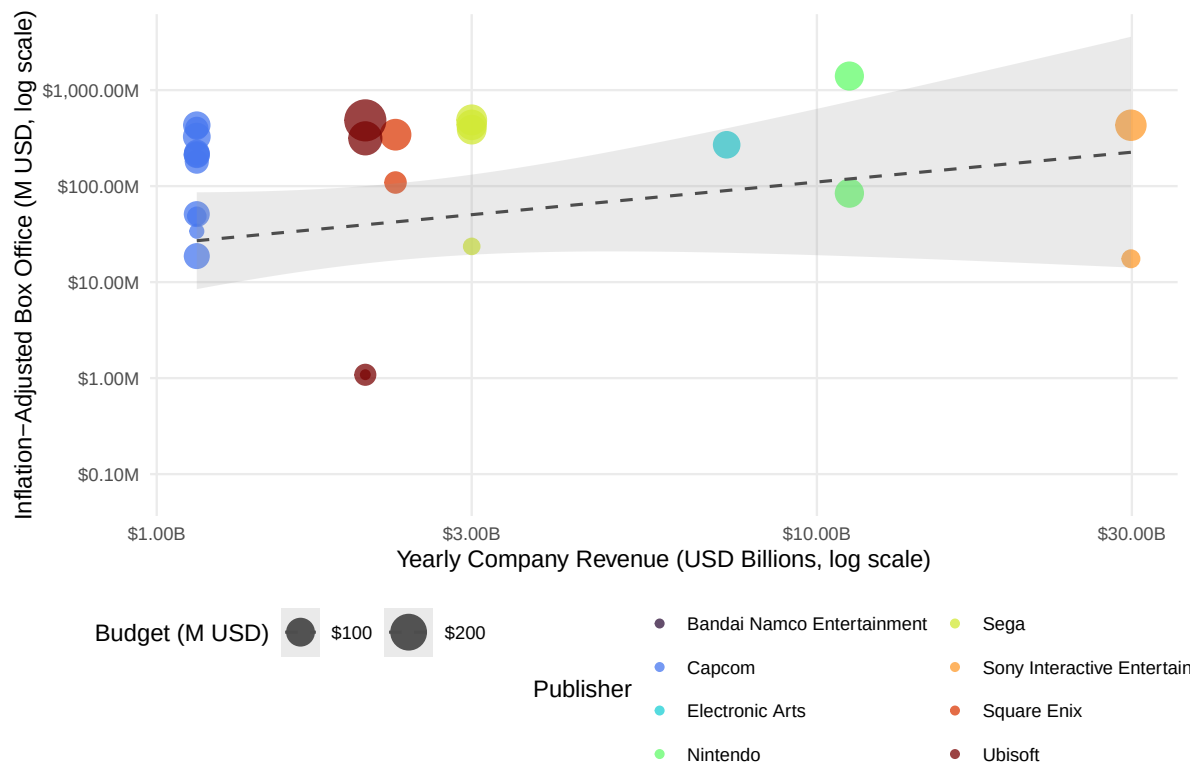


Figure 3: Publisher revenue vs. box office performance (3+ variable figure). Color = publisher, point size = budget, x = company annual revenue, y = inflation-adjusted box office.

publishers by revenue, most publishers cluster in a relatively narrow band of box office performance regardless of their annual revenue. The substantial overlap in earnings ranges across publishers is more clearly visualized in Figure 4, a ridgeline density plot showing the full distribution of box office outcomes for each publisher.

Picking joint bandwidth of 0.543

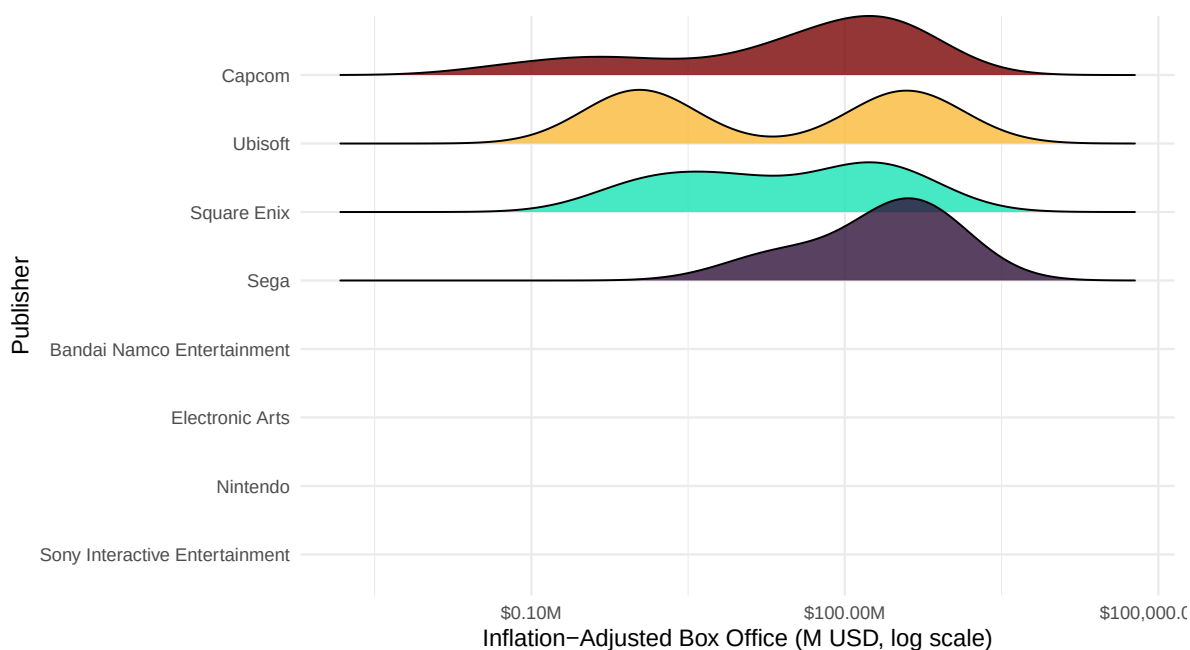


Figure 4: Distribution of inflation-adjusted box office earnings by publisher. Ridgeline density plot on a log scale.

Several patterns are immediately visible. Nintendo's distribution shows a massive rightward skew driven by the *Super Mario* films, but also includes lower-earning adaptations. Capcom, Square Enix, and Sega show multi-modal distributions reflecting the variable performance of their respective film franchises. The wide overlap in earnings distributions across publishers of vastly different sizes reinforces the conclusion from Figure 3.

Key findings for Q2: Publisher size, as measured by annual corporate revenue, is a weak predictor of box office success for video game film adaptations. While the largest publishers occasionally produce outsized hits (notably Nintendo's *Super Mario* films), smaller publishers like Capcom and Square Enix have achieved comparable box office results. The relationship between publisher resources and box office performance appears to be mediated more by the specific intellectual property and creative execution than by the raw financial scale of the publishing company.

Q3: Franchise Fatigue

Our final research question investigates whether video game film franchises exhibit “franchise fatigue” — the tendency for later installments in a film series to earn less at the box office and receive worse critical reception than earlier entries. This pattern is well-documented in traditional Hollywood franchises (*Fast & Furious*, *Transformers*, etc.) but has not been systematically studied for video game adaptations.

To identify franchises within our dataset, we extracted the first meaningful word from each film’s title (removing leading articles “The,” “A,” and “An”) and grouped films sharing the same first word. We assigned installment numbers within each franchise by chronological order of release. To focus on substantive franchises with meaningful trends, we retained only those with three or more theatrical-release films. This yielded a set of major franchises including *Resident Evil*, *Pokémon*, *Tomb Raider*, *Mortal Kombat*, and several others.

Figure 5 plots inflation-adjusted box office by installment number for each major franchise, with individual trend lines per franchise and an overall linear-model trend (dashed line).

```
`geom_smooth()` using formula = 'y ~ x'
```

Figure 5 reveals a mixed but suggestive picture. The overall linear-model trend (dashed line) shows a gentle downward slope, consistent with franchise fatigue. However, individual franchises tell more nuanced stories. The *Resident Evil* series — the most prolific franchise in our dataset with six theatrical films — shows a clear downward trajectory across its original run. *Pokémon* films exhibit more variable performance, with earlier installments performing strongly before a decline in later entries. *Tomb Raider* shows only two theatrical films in our filtered dataset (after removing direct-to-video entries), limiting trend analysis for that franchise.

The companion visualization in Figure 6 examines whether critic scores (Rotten Tomatoes) follow the same pattern of decline across installments.

```
`geom_smooth()` using formula = 'y ~ x'
```

Critic scores show a clearer downward trend than box office figures, with the overall linear model sloping more decisively downward. This suggests that even when later franchise installments manage to maintain or grow their box office (perhaps through built-in audience loyalty or nostalgia), professional critics become increasingly harsh on successive entries. The *Resident Evil* franchise again exemplifies this pattern, with critics scoring later installments substantially lower than the original.

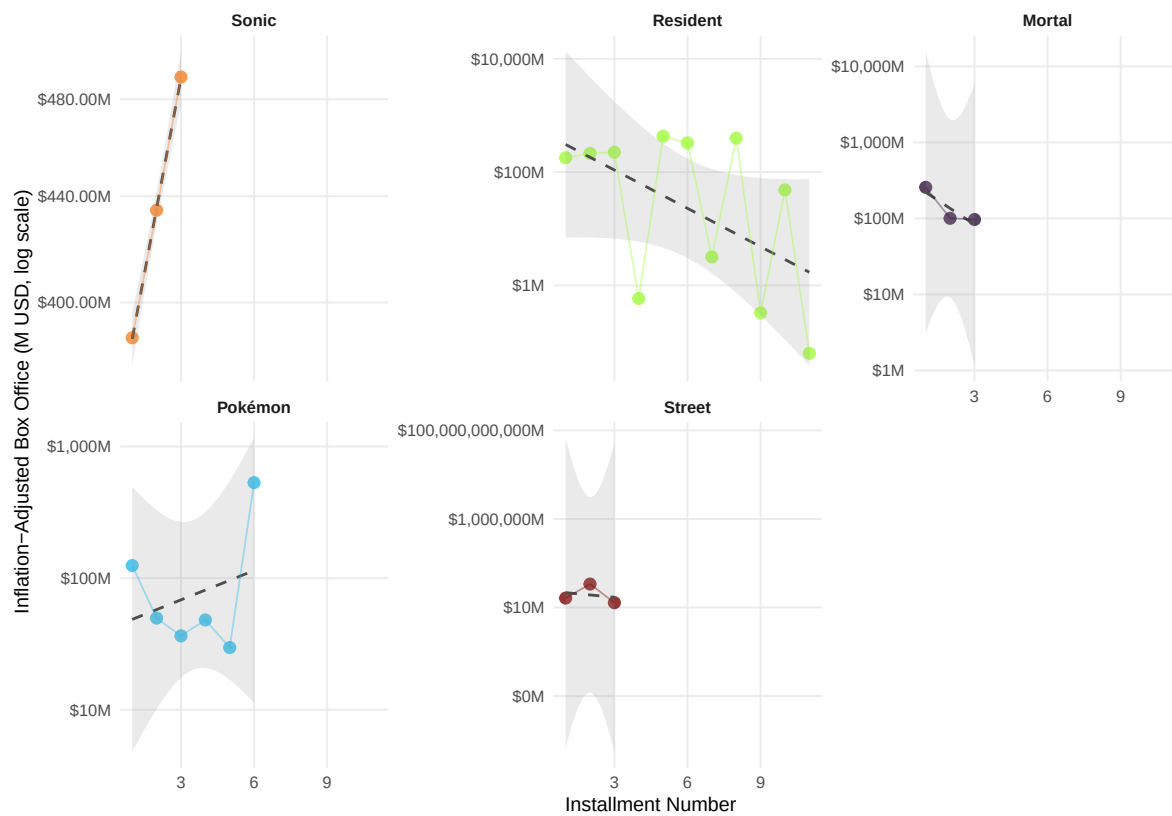


Figure 5: Inflation-adjusted box office by installment number for major video game film franchises (3+ films). Faceted by franchise with log-scale y-axis.

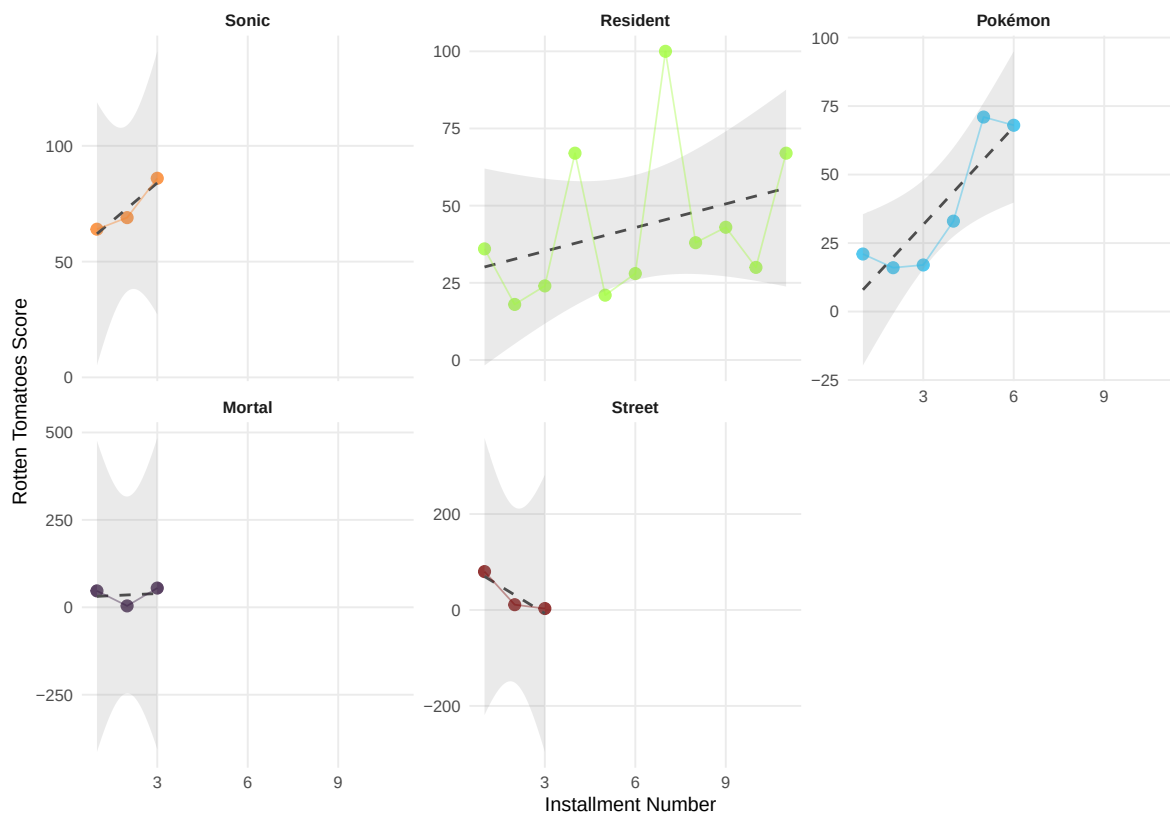


Figure 6: Rotten Tomatoes critic score by installment number for major video game film franchises.

Figure 7 provides a complementary view — a tile heatmap where color intensity represents box office magnitude, allowing direct visual comparison of franchise performance across installments.

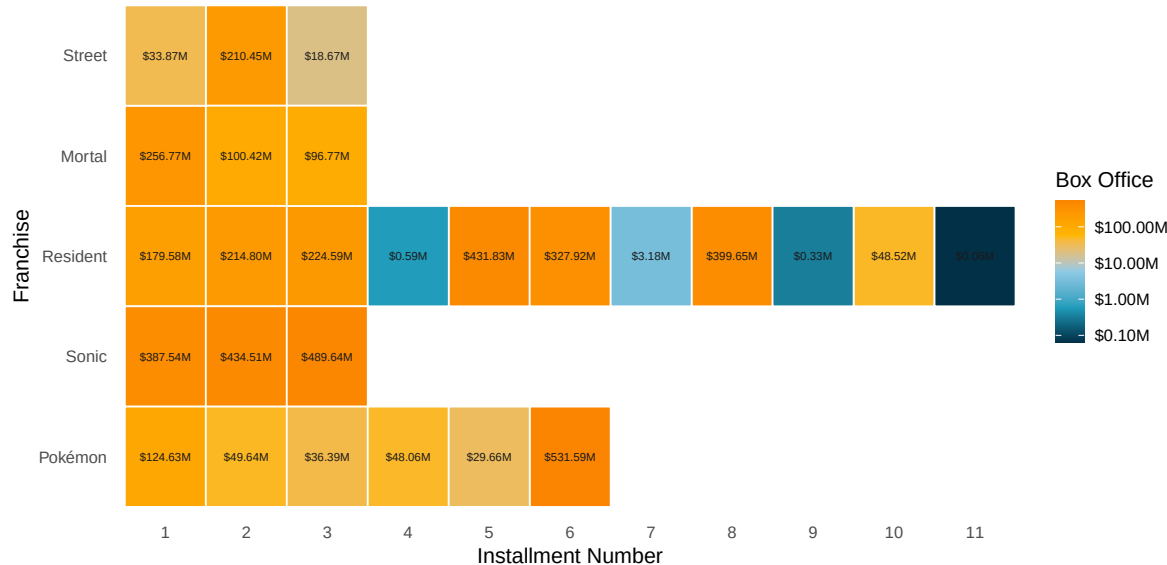


Figure 7: Franchise $\times$ installment box office heatmap. Color intensity represents inflation-adjusted box office on a log scale.

The heatmap in Figure 7 makes the franchise fatigue pattern immediately visible through color gradients: for franchises like *Resident Evil* and *Silent Hill*, the color intensity fades from left (early installments) to right (later installments), indicating declining box office performance over time. Other franchises, such as *Hitman* and *Rampage*, have too few theatrical entries to display a meaningful pattern.

Key findings for Q3: Franchise fatigue appears to be a real phenomenon in video game film adaptations, though its manifestation varies by franchise and metric. Box office shows a modest overall decline across successive installments, while critical reception declines more consistently and steeply. The most prolific franchise in the dataset (*Resident Evil*, six theatrical films) exhibits clear fatigue across both financial and critical dimensions. These findings suggest that studios developing video game film franchises should anticipate diminishing returns and consider strategies — such as creative reboots, extended gaps between installments, or higher production values — to counteract franchise fatigue.

Correlation Analysis

To complement our question-specific analyses, Figure 8 presents a correlation matrix of all key numeric variables in the dataset. This visualization uses a diverging red-white-teal color scale (red = negative correlation, teal = positive) with circle sizes proportional to correlation strength.

```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.  
i Please use tidy evaluation idioms with `aes()``.  
i See also `vignette("ggplot2-in-packages")` for more information.  
i The deprecated feature was likely used in the ggcorrplot package.  
Please report the issue at <https://github.com/kassambara/ggcorrplot/issues>.
```

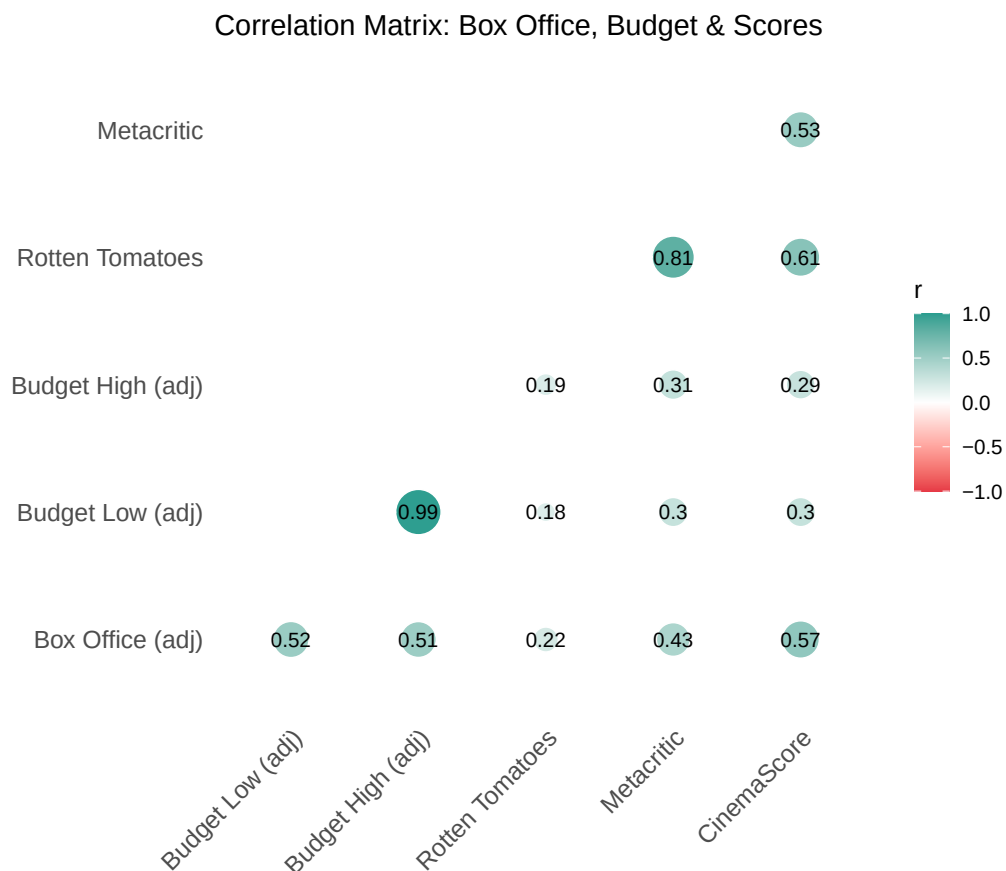


Figure 8: Correlation matrix of box office, budget, and review score variables. Diverging red-white-teal color scale with circle areas proportional to correlation magnitude.

Several notable relationships emerge from Figure 8. Budget and box office are moderately positively correlated ($r = 0.4\text{--}0.5$), which aligns with the intuition that more expensive films tend to earn more — though the correlation is far from perfect, indicating that high budgets do not guarantee high returns. The two critic score platforms, Rotten Tomatoes and Metacritic, are strongly correlated ($r = 0.85$), as expected given that they both aggregate professional critic reviews (albeit using different methodologies). CinemaScore (audience) shows a modest positive correlation with both critic metrics ($r = 0.3\text{--}0.4$), consistent with our Q1 finding that audiences and critics generally move in the same direction but at different absolute levels. Most notably, budget shows the weakest correlation with CinemaScore ($r = 0.11$), suggesting that higher production spending does not meaningfully improve how audiences rate these films.

Findings / Conclusions

This project investigated three questions about video game film adaptations using a dataset spanning from 1986 to 2024. Our key findings are:

Q1 — Critic–Audience Agreement: Audiences consistently rate video game films higher than professional critics, with a persistent gap of approximately 10–20 points on a 0–100 scale. The gap has narrowed modestly since 2010, and all score types show a gentle upward trend over the same period. This suggests that while the genre still faces critical skepticism, both the quality of films and the alignment between critic and audience expectations may be improving.

Q2 — Publisher Size and Box Office: Contrary to what one might expect, the annual revenue of a game’s publishing company is a weak predictor of its film adaptation’s box office performance. While the largest publishers occasionally produce breakout hits (Nintendo’s *Super Mario* films), smaller publishers have achieved comparable financial results. The specific intellectual property and the creative team behind each film appear to matter more than the corporate scale of the publisher.

Q3 — Franchise Fatigue: Video game film franchises do exhibit signs of franchise fatigue, particularly in critical reception, which declines more steeply across installments than box office returns. The most prolific franchise (*Resident Evil*, with six theatrical entries) demonstrates clear fatigue in both financial and critical dimensions. Studios pursuing franchise strategies should anticipate diminishing returns with each successive installment.

Taken together, these findings paint a picture of a genre in transition. Video game film adaptations have historically been received coolly by critics but warmly by audiences, a gap that may be narrowing as the industry matures. Financial success appears to depend more on the specific property and execution than on the resources of the publisher. And while franchise-building is a common strategy, the returns do diminish — suggesting that quality and novelty, rather than volume, may be the key to sustained success in this space.

Limitations & Future Work

Several limitations should be considered when interpreting these findings:

- **Exchange Rates:** Our currency standardization used fixed approximate exchange rates. Historical daily rates would improve the accuracy of non-USD conversions, particularly for films from Japan and Hong Kong where exchange rates have fluctuated significantly over the decades covered by our dataset.
- **CPI Data:** We used annual CPI-U averages. For films released early or late in a calendar year, a more precise monthly adjustment could refine the inflation correction.
- **Missing Data:** A substantial number of films in the original dataset lack box office data, critic scores, or both — particularly older and smaller-scale releases. Our analyses are necessarily restricted to films with complete data, which may bias results toward larger, more prominent releases that are more likely to have been documented comprehensively.
- **Publisher Revenue Matching:** Only a subset of game publishers in the film dataset matched the Wikipedia list of largest companies by revenue. Many smaller or defunct publishers were excluded from Q2, limiting the generalizability of our publisher-size findings. A more comprehensive source of publisher financial data would strengthen this analysis.
- **Franchise Detection:** Our first-word heuristic for franchise detection is imperfect. It may group films incorrectly (e.g., franchises with multi-word names that share a common first word with unrelated films) and does not capture subtitled sequels where the first word differs (e.g., “The Matrix” vs. “The Animatrix”). A more sophisticated approach using regular expressions or manual franchise coding would improve accuracy.
- **Causality:** All analyses in this project are correlational. We cannot determine, for example, whether publisher size *causes* higher box office or whether a few successful films inflate publisher revenue figures.

Future work could extend this analysis in several directions: incorporating genre and rating data to control for film characteristics; developing a predictive model for box office performance using publisher, budget, and critic variables; analyzing the impact of simultaneous vs. staggered global release strategies; or examining whether the relationship between video game release dates and film adaptation release dates affects box office outcomes.

AI Use Statement

Harsh Rathi: [Describe how you used AI tools in your contributions]

Charles Bupp: [Describe how you used AI tools in your contributions]

Mason Mitchell: [Describe how you used AI tools in your contributions]

References

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- R Core Team (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.
- Wickham, H., et al. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686.

Code Appendix

Chunk setup:

```
library(tidyverse)
library(readr)
library(janitor)
library(scales)
library(rvest)
library(lubridate)
library(ggcorrplot)
library(patchwork)
library(ggthemes)
library(kableExtra)
library(viridis)

ggplot2::theme_set(theme_minimal(base_size = 12))
```

Chunk provenance-load:

```
game_films <- readr::read_csv(
  "https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2026/2026-06-09/game_films.csv"
) |>
janitor::clean_names()
```

Chunk currency-table:

```
fx_rates <- tibble::tribble(
  ~currency, ~to_usd,
  "$",      1,
  "¥",      0.0091,
  "£",      1.30,
  "HK$",    0.128,
  "NT$",    0.033
)

fx_rates |>
  rename(Currency = currency, `USD per Unit` = to_usd) |>
  kable(caption = "Exchange rates used for currency standardization", digits = 4) |>
  kable_styling(bootstrap_options = c("striped", "condensed"), full_width = FALSE)
```

Chunk currency-convert:

```
game_films <- game_films |>
  left_join(fx_rates, by = c("worldwide_box_office_currency" = "currency")) |>
  mutate(worldwide_box_office_usd = worldwide_box_office * to_usd) |>
  select(-to_usd) |>
  left_join(fx_rates, by = c("budget_currency" = "currency")) |>
  mutate(
    budget_low_usd = budget_low * to_usd,
    budget_high_usd = budget_high * to_usd
  ) |>
  select(-to_usd)
```

Chunk cpi-data:

```
cpi <- tibble::tribble(
  ~year, ~cpi,
  1986, 109.6, 1987, 113.6, 1988, 118.3, 1989, 124.0,
  1990, 130.7, 1991, 136.2, 1992, 140.3, 1993, 144.5,
  1994, 148.2, 1995, 152.4, 1996, 156.9, 1997, 160.5,
  1998, 163.0, 1999, 166.6, 2000, 172.2, 2001, 177.1,
  2002, 179.9, 2003, 184.0, 2004, 188.9, 2005, 195.3,
  2006, 201.6, 2007, 207.3, 2008, 215.3, 2009, 214.5,
  2010, 218.1, 2011, 224.9, 2012, 229.6, 2013, 233.0,
  2014, 236.7, 2015, 237.0, 2016, 240.0, 2017, 245.1,
  2018, 251.1, 2019, 255.7, 2020, 258.8, 2021, 271.0,
```

```

    2022, 292.7, 2023, 304.7, 2024, 313.7
  )

cpi_2024 <- cpi$cpi[cpi$year == 2024]

game_films <- game_films |>
  mutate(release_year = lubridate::year(release_date)) |>
  left_join(cpi, by = c("release_year" = "year")) |>
  mutate(
    inflation_factor = cpi_2024 / cpi,
    worldwide_box_office_usd_adj = worldwide_box_office_usd * inflation_factor,
    budget_low_usd_adj = budget_low_usd * inflation_factor,
    budget_high_usd_adj = budget_high_usd * inflation_factor
  ) |>
  select(-cpi, -inflation_factor)

```

Chunk theatrical-filter:

```

game_films <- game_films %>%
  filter(category == "Theatrical releases")

```

Chunk cinema-convert:

```

cinema_score_conversion <- tibble::tribble(
  ~cinemascore_letter, ~to_num,
  "N/A", NA,
  "A+", 100, "A", 91.6666, "A-", 83.3333,
  "B+", 74.9999, "B", 66.6666, "B-", 58.3333,
  "C+", 49.9999, "C", 41.6666, "C-", 33.3333,
  "D+", 24.9999, "D", 16.6666, "D-", 8.3333,
  "F", 0
)

game_films <- game_films %>%
  left_join(cinema_score_conversion, join_by("cinema_score" == "cinemascore_letter"))

game_films <- game_films %>%
  mutate(cinema_score = to_num) %>%
  select(-starts_with("to_num"))

```

Chunk final-summary:

```

game_films |>
  summarise(
    `Total Films` = n(),
    `Films with Box Office Data` = sum(!is.na(worldwide_box_office_usd_adj)),
    `Films with RT Score` = sum(!is.na(rotten_tomatoes)),
    `Films with Metacritic` = sum(!is.na(metacritic)),
    `Films with CinemaScore` = sum(!is.na(cinema_score)),
    `Date Range` = paste(range(release_date, na.rm = TRUE), collapse = " to ")
  ) |>
  kable(
    caption = "Overview of the cleaned dataset after all preparation steps",
    digits = 0,
    align = "l"
  ) |>
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE)

```

Chunk tbl-summary-stats:

```

game_films |>
  select(
    worldwide_box_office_usd_adj,
    budget_high_usd_adj,
    rotten_tomatoes,
    metacritic,
    cinema_score
  ) |>
  summarise(across(everything(), list(
    Mean = ~mean(.x, na.rm = TRUE),
    Median = ~median(.x, na.rm = TRUE),
    SD = ~sd(.x, na.rm = TRUE),
    Min = ~min(.x, na.rm = TRUE),
    Max = ~max(.x, na.rm = TRUE)
  ))) |>
  pivot_longer(everything(), names_sep = "_", names_to = c("Variable", "Statistic")) |>
  pivot_wider(names_from = Statistic, values_from = value) |>
  mutate(Variable = recode(Variable,
    worldwide_box_office_usd_adj = "Box Office (adj. USD)",
    budget_high_usd_adj = "Budget High (adj. USD)",
    rotten_tomatoes = "Rotten Tomatoes",
    metacritic = "Metacritic",
    cinema_score = "CinemaScore"
  )) |>

```

```
kable(digits = 1, align = c("l", rep("r", 5))) |>
kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE)
row_spec(0, bold = TRUE)
```

Chunk fig-q1-scatter:

```
aud_v_cri <- game_films %>%
  filter(!is.na(rotten_tomatoes) | !is.na(metacritic), !is.na(cinema_score)) %>%
  pivot_longer(cols = rotten_tomatoes:cinema_score, names_to = "critic_type", values_to = "score")
  select(title, critic_type, score, release_date)

critic_colors <- c(
  "rotten_tomatoes" = "#E63946",
  "metacritic"      = "#457B9D",
  "cinema_score"    = "#2A9D8F"
)

critic_labels <- c(
  "rotten_tomatoes" = "Rotten Tomatoes (Critic)",
  "metacritic"      = "Metacritic (Critic)",
  "cinema_score"    = "CinemaScore (Audience)"
)

aud_v_cri %>%
  ggplot(aes(x = release_date, y = score, color = critic_type)) +
  geom_point(alpha = 0.5, size = 2.5) +
  geom_smooth(se = TRUE, method = "loess", span = 0.6, alpha = 0.15) +
  scale_color_manual(values = critic_colors, labels = critic_labels) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  labs(x = "Release Date", y = "Score", color = "Score Type") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())
```

Chunk fig-q1-gap:

```
aud_v_cri_gap <- game_films %>%
  filter(!is.na(rotten_tomatoes) | !is.na(metacritic), !is.na(cinema_score)) %>%
  mutate(
    avg_critic = rowMeans(cbind(rotten_tomatoes, metacritic), na.rm = TRUE),
    gap = avg_critic - cinema_score
  ) %>%
```

```

filter(!is.na(gap))

aud_v_cri_gap %>%
  ggplot(aes(x = release_date, y = gap)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray40", linewidth = 0.8) +
  geom_point(aes(fill = gap > 0), size = 2.5, alpha = 0.7, shape = 21, color = "white") +
  geom_smooth(method = "loess", se = TRUE, color = "#E63946", fill = "#E63946", alpha = 0.15) +
  scale_fill_manual(
    values = c("TRUE" = "#2A9D8F", "FALSE" = "#E76F51"),
    labels = c("TRUE" = "Critics Higher", "FALSE" = "Audience Higher")
  ) +
  scale_x_date(date_breaks = "5 years", date_labels = "%Y") +
  labs(x = "Release Date", y = "Gap (Critic - Audience)", fill = "Direction") +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())

```

Chunk tbl-publisher-performance:

```

# Scrape and join publisher data
company_url <- "https://en.wikipedia.org/wiki/List_of_largest_video_game_companies_by_revenue"
TablesRedun <- company_url |>
  read_html() |>
  html_elements("table") |>
  html_table()
top_50 <- TablesRedun[[2]]

top_50_clean <- clean_names(top_50) %>%
  mutate(revenue_usd_billions = parse_number(revenue_usd)) %>%
  select(!c(ref, revenue_usd))

pub_joined <- as.data.frame(top_50_clean) %>%
  right_join(game_films, join_by("company" == "original_game_publisher"), copy)

table_data <- pub_joined %>%
  filter(!is.na(worldwide_box_office_usd_adj)) %>%
  group_by(company) %>%
  summarise(
    `# Films` = n(),
    `Median Box Office (M USD)` = round(median(worldwide_box_office_usd_adj) / 1e6, 1),
    `Mean Box Office (M USD)` = round(mean(worldwide_box_office_usd_adj) / 1e6, 1),
    `Mean RT Score` = round(mean(rotten_tomatoes, na.rm = TRUE), 1),
    `Mean CinemaScore` = round(mean(cinema_score, na.rm = TRUE), 1),
  )

```

```

    .groups = "drop"
  ) %>%
  filter(`# Films` >= 3) %>%
  arrange(desc(`Mean Box Office (M USD)`))

table_data %>%
  kable(digits = 1, align = c("l", rep("r", 5))) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE) %>%
  row_spec(0, bold = TRUE, color = "white", background = "#2A9D8F")

```

Chunk fig-q2-multivariate:

```

pub_joined %>%
  filter(!is.na(worldwide_box_office_usd_adj), !is.na(revenue_usd_billions)) %>%
  ggplot(aes(x = revenue_usd_billions,
             y = worldwide_box_office_usd_adj / 1e6,
             color = company,
             size = budget_high_usd_adj / 1e6)) +
  geom_point(alpha = 0.75) +
  geom_smooth(aes(group = 1), method = "lm", se = TRUE, color = "gray30",
             linewidth = 0.8, linetype = "dashed", alpha = 0.2) +
  scale_color_viridis_d(option = "turbo", guide = guide_legend(ncol = 2)) +
  scale_size_continuous(name = "Budget (M USD)", labels = dollar_format(), range = c(2, 10)) +
  scale_x_log10(labels = dollar_format(suffix = "B")) +
  scale_y_log10(labels = dollar_format(suffix = "M")) +
  labs(
    x = "Yearly Company Revenue (USD Billions, log scale)",
    y = "Inflation-Adjusted Box Office (M USD, log scale)",
    color = "Publisher"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "bottom", panel.grid.minor = element_blank())

```

Chunk fig-q2-ridge:

```

pub_joined %>%
  filter(!is.na(worldwide_box_office_usd_adj), !is.na(revenue_usd_billions)) %>%
  mutate(company = fct_reorder(company, revenue_usd_billions, .desc = TRUE)) %>%
  ggplot(aes(x = worldwide_box_office_usd_adj / 1e6, y = company, fill = company)) +
  geom_density_ridges(alpha = 0.8, scale = 1.2) +
  scale_fill_viridis_d(option = "turbo", guide = "none") +

```

```
scale_x_log10(labels = dollar_format(suffix = "M")) +
labs(x = "Inflation-Adjusted Box Office (M USD, log scale)", y = "Publisher") +
theme_minimal(base_size = 13)
```

Chunk fig-q3-box-office:

```
MovieCount <- game_films %>%
  mutate(
    title_clean = str_remove(title, "^(The|A|An) "),
    start_word = str_extract(title_clean, "^[^ ]+")
  ) %>%
  select(title, start_word, release_date, worldwide_box_office_usd_adj, rotten_tomatoes) %>%
  filter(if_all(everything(), \(x) !is.na(x)))

MovieCountSummary <- MovieCount %>%
  group_by(start_word) %>%
  count() %>%
  filter(n >= 3) %>%
  arrange(desc(n))

MovieCountFinal <- semi_join(MovieCount, MovieCountSummary, join_by(start_word)) %>%
  group_by(start_word) %>%
  mutate(installment = rank(release_date)) %>%
  ungroup()

MovieCountFinal %>%
  ggplot(aes(x = installment, y = worldwide_box_office_usd_adj / 1e6)) +
  geom_point(aes(color = start_word), size = 3, alpha = 0.8) +
  geom_smooth(aes(group = 1), method = "lm", se = TRUE, color = "gray30",
    linetype = "dashed", linewidth = 0.8, alpha = 0.2) +
  geom_line(aes(group = start_word, color = start_word), linewidth = 0.5, alpha = 0.4) +
  scale_color_viridis_d(option = "turbo", guide = "none") +
  scale_y_log10(labels = dollar_format(suffix = "M")) +
  facet_wrap(~ reorder(start_word, -worldwide_box_office_usd_adj), scales = "free_y") +
  labs(x = "Installment Number", y = "Inflation-Adjusted Box Office (M USD, log scale)") +
  theme_minimal(base_size = 12) +
  theme(strip.text = element_text(face = "bold"), panel.grid.minor = element_blank())
```

Chunk fig-q3-rt:

```

MovieCountFinal %>%
  ggplot(aes(x = installment, y = rotten_tomatoes)) +
  geom_point(aes(color = start_word), size = 3, alpha = 0.8) +
  geom_smooth(aes(group = 1), method = "lm", se = TRUE, color = "gray30",
              linetype = "dashed", linewidth = 0.8, alpha = 0.2) +
  geom_line(aes(group = start_word, color = start_word), linewidth = 0.5, alpha = 0.4) +
  scale_color_viridis_d(option = "turbo", guide = "none") +
  facet_wrap(~ reorder(start_word, -rotten_tomatoes), scales = "free_y") +
  labs(x = "Installment Number", y = "Rotten Tomatoes Score") +
  theme_minimal(base_size = 12) +
  theme(strip.text = element_text(face = "bold"), panel.grid.minor = element_blank())

```

Chunk fig-q3-heatmap:

```

MovieCountFinal %>%
  mutate(start_word = fct_reorder(start_word, worldwide_box_office_usd_adj, .fun = max, .desc = TRUE)) +
  ggplot(aes(x = factor(installment), y = start_word, fill = worldwide_box_office_usd_adj / 1e6)) +
  geom_tile(color = "white", linewidth = 0.5) +
  geom_text(aes(label = scales::dollar(worldwide_box_office_usd_adj / 1e6, suffix = "M")),
            size = 2.5, color = "gray10") +
  scale_fill_gradientn(
    colors = c("#023047", "#219EBC", "#8ECAE6", "#FFB703", "#FB8500"),
    labels = dollar_format(suffix = "M"),
    na.value = "gray90",
    trans = "log10"
  ) +
  labs(x = "Installment Number", y = "Franchise", fill = "Box Office") +
  theme_minimal(base_size = 13) +
  theme(panel.grid = element_blank(), axis.ticks = element_blank())

```

Chunk fig-correlation:

```

cor_vars <- game_films %>%
  select(
    worldwide_box_office_usd_adj,
    budget_low_usd_adj,
    budget_high_usd_adj,
    rotten_tomatoes,
    metacritic,
    cinema_score
  ) %>%

```

```

rename(
  `Box Office (adj)` = worldwide_box_office_usd_adj,
  `Budget Low (adj)` = budget_low_usd_adj,
  `Budget High (adj)` = budget_high_usd_adj,
  `Rotten Tomatoes` = rotten_tomatoes,
  `Metacritic` = metacritic,
  `CinemaScore` = cinema_score
)

cor_matrix <- cor(cor_vars, use = "pairwise.complete.obs")

ggcorrplot(
  cor_matrix,
  method = "circle",
  type = "lower",
  lab = TRUE,
  lab_size = 3.5,
  colors = c("#E63946", "white", "#2A9D8F"),
  title = "Correlation Matrix: Box Office, Budget & Scores",
  ggtheme = theme_minimal(base_size = 12),
  legend.title = "r",
  outline.color = "white"
) +
  theme(panel.grid.major = element_blank())

```